

Phase B — Persistence Diagram Metrics (Wind MR→HR)

Goal: compare SR outputs (GAN vs CNN) against GT using PSNR + PD distances.

Baseline: bicubic upsampling from MR input.

Configuration:

- Dataset: example_data/wind_MR-HR.tfrecord (5 records) -> 5 images per run
- Runs: CNN_MRHR, GAN_MRHR
- Field(s): speed
- Patch/stride: patch=160, stride=160
- PD pruning: min_pers=0.01, max_pts=120
- W2 audit: w2_audit_every=3 (~1/K patches)
- Rows: 90 (patch-level), summarized per-sample in stats CSV

Notes:

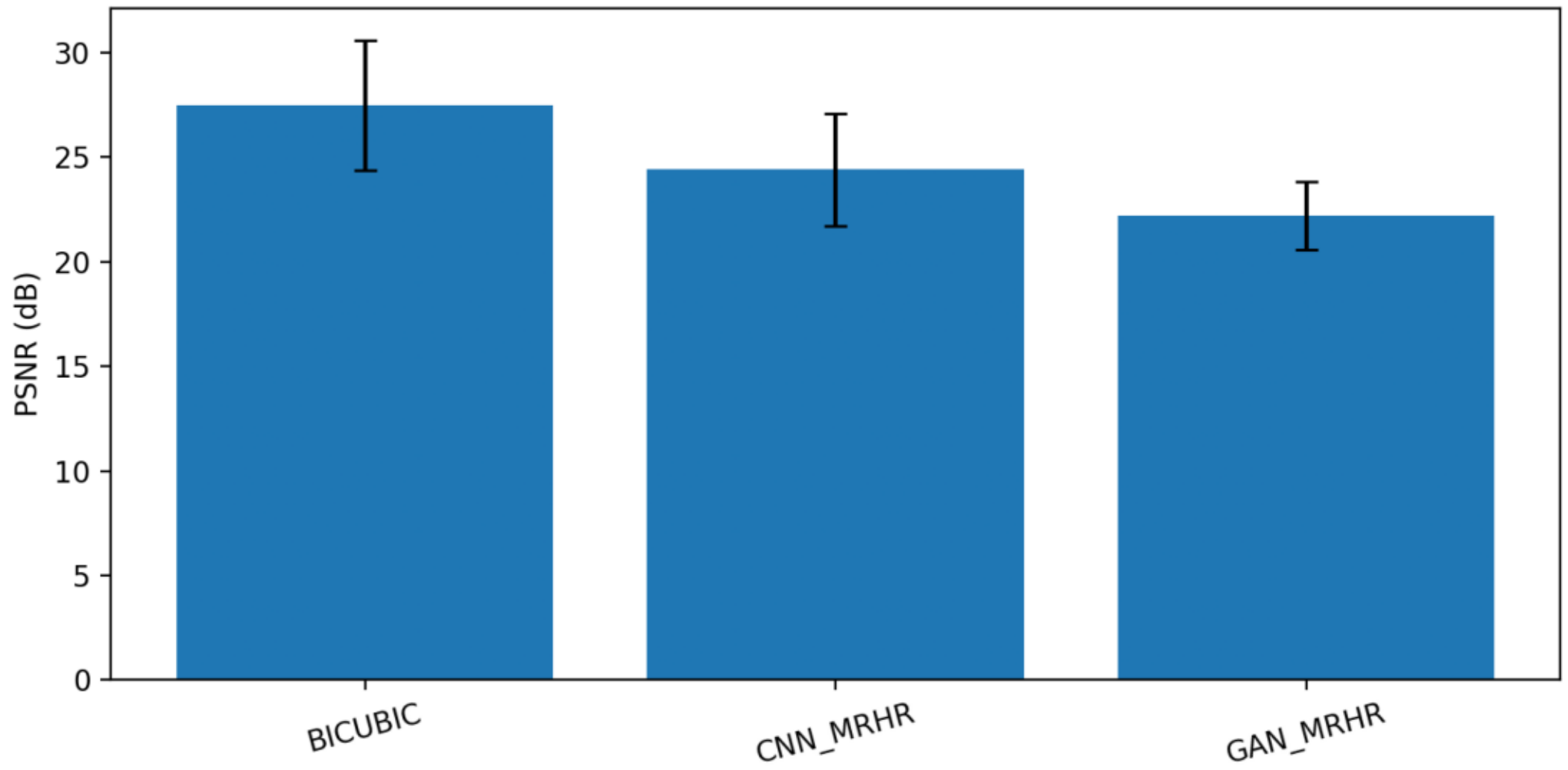
- PD computed on speed field ($||u,v||$) in $[0,1]$ (pairwise normalized by GT min/max).
- W1 computed for all patches; W2 computed on a deterministic audit subset to keep runtime reasonable

Summary statistics (per-sample avg of patches)

	label	n_samples	patches_per_sample	psnr_mean	psnr_std	pd0_mean	pd0_std	pd1_mean	pd1_std	pd2_mean	pd2_std	pd3_mean	pd3_std	pd0_count	pd1_count
	BICUBIC	5	45	27.51	3.108	6.647	1.55	7.515	2.879	0.7022	0.1255			12	15
CNN_MRH		5	45	24.42	2.68	5.225	1.015	6.248	2.115	0.6043	0.1309			12	15
GAN_MRH		5	45	22.2	1.619	5.303	1.464	4.602	1.035	0.5199	0.1143	0.4784	0.07161	13	16

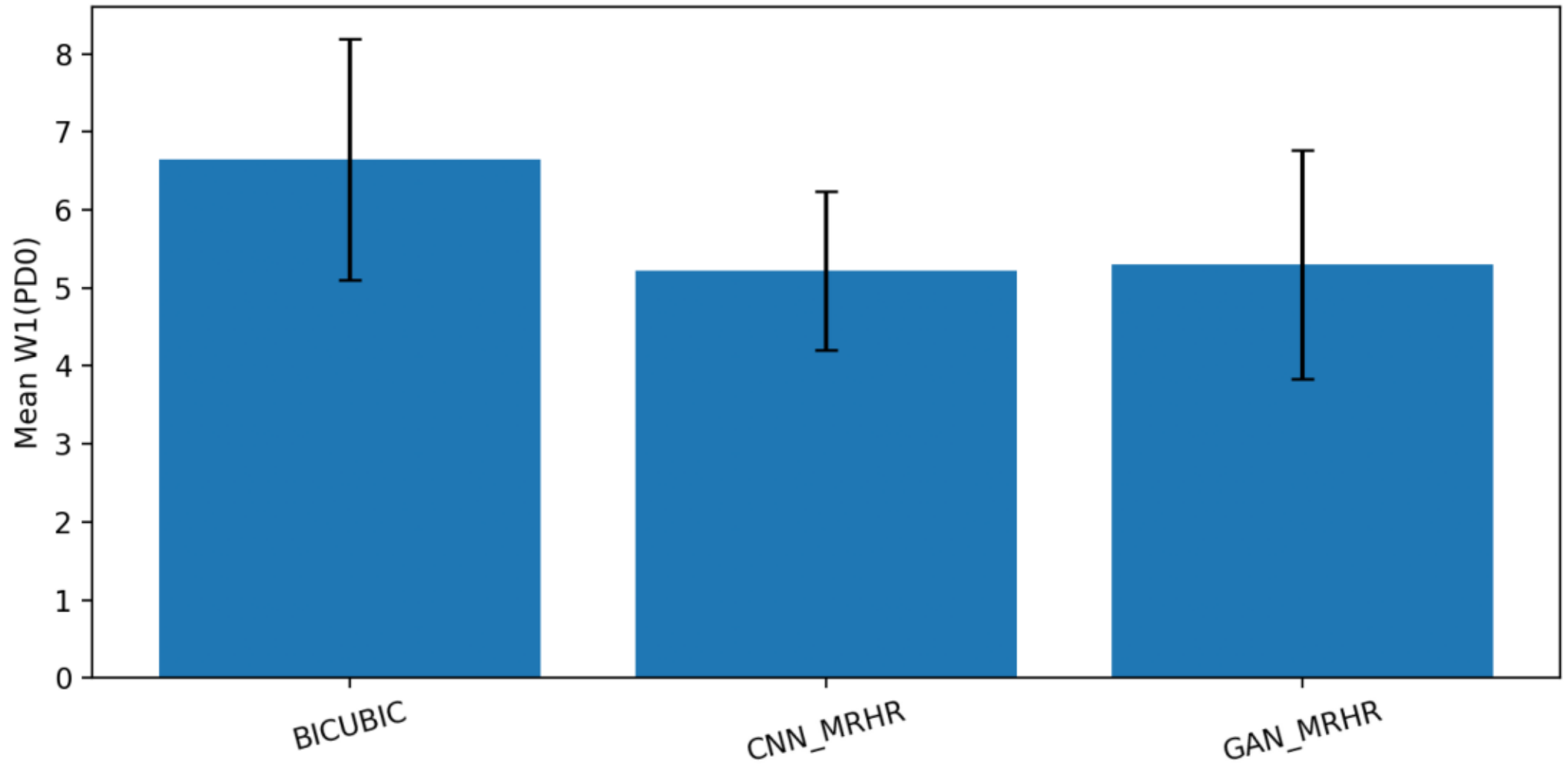
Mean PSNR (higher better)

Mean PSNR vs GT on speed (per-sample avg of patches)



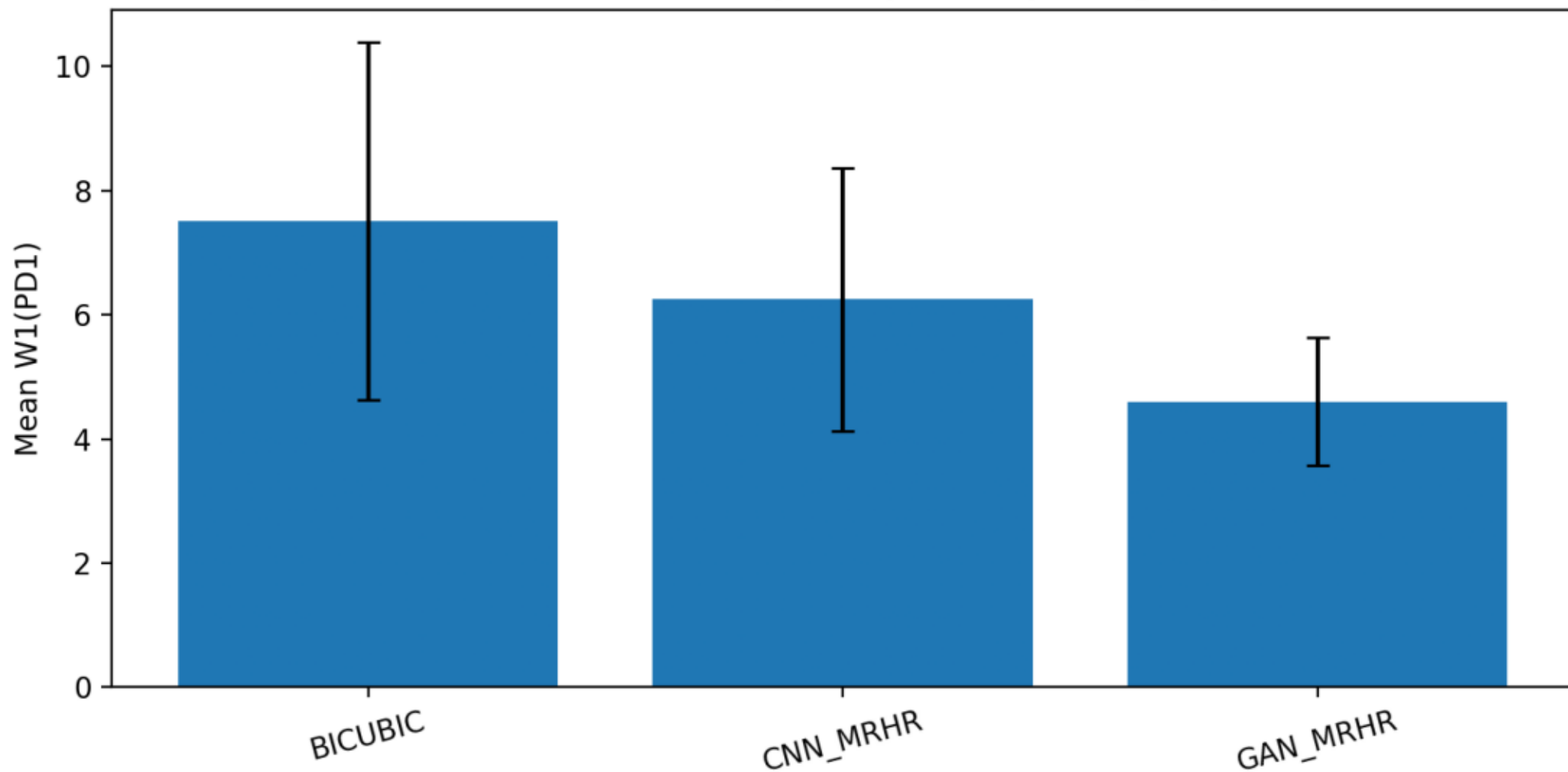
Mean W1(PD0) (lower better)

Mean W1(PD0) vs GT on speed (per-sample avg of patches)



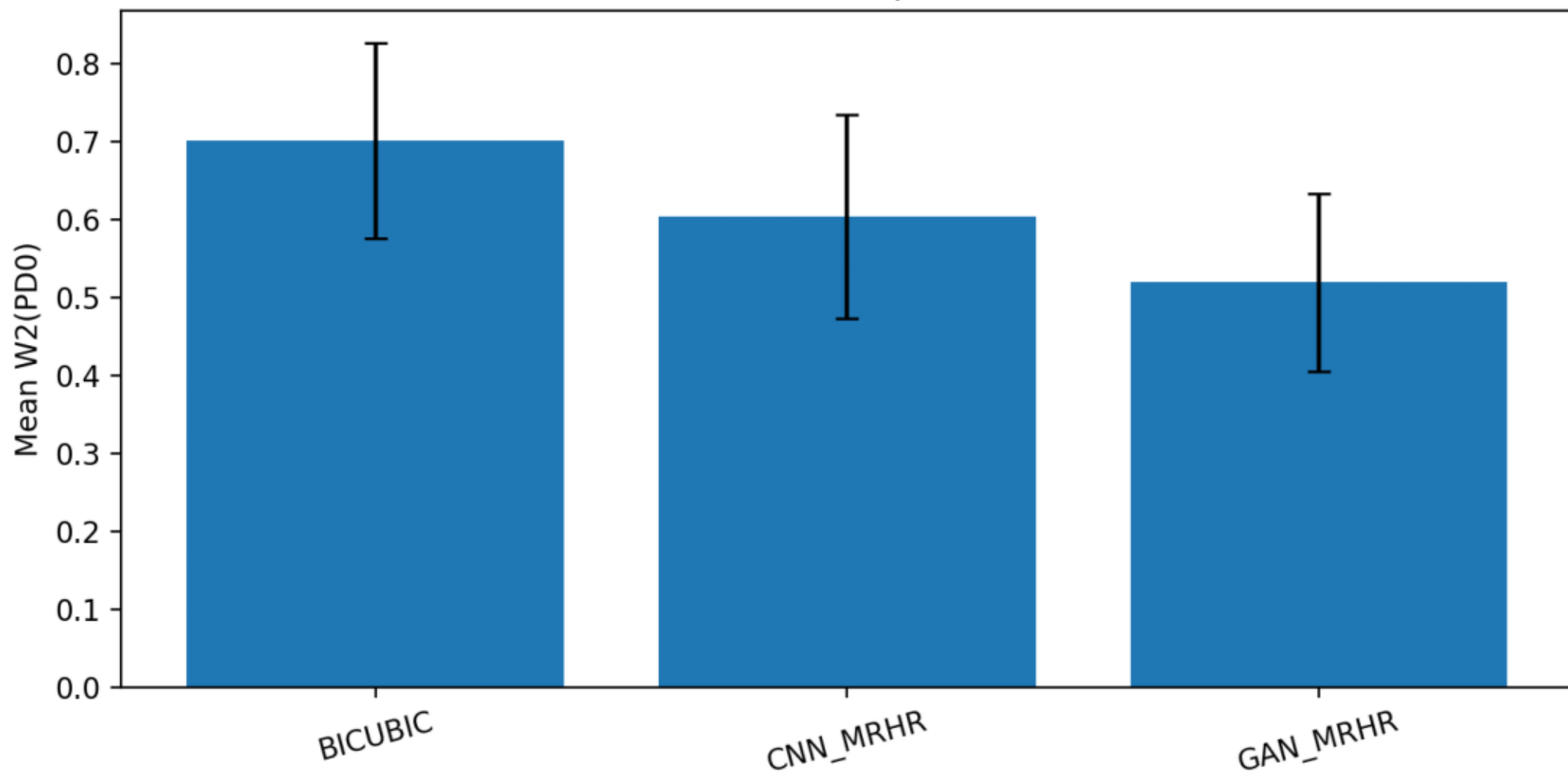
Mean W1(PD1) (lower better)

Mean W1(PD1) vs GT on speed (per-sample avg of patches)



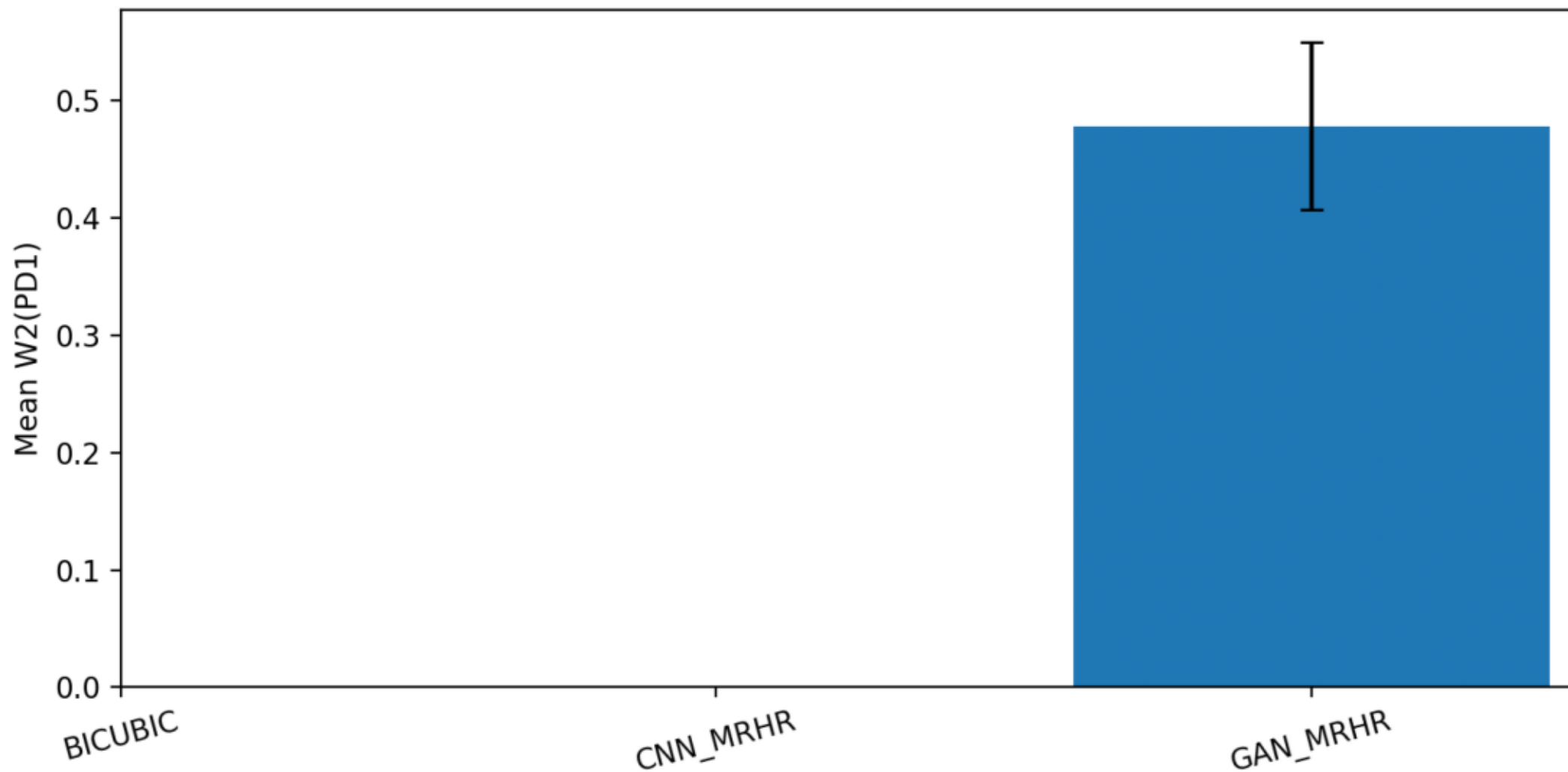
Mean W2(PD0) audit subset

Mean W2(PD0) vs GT on speed (AUDIT subset)



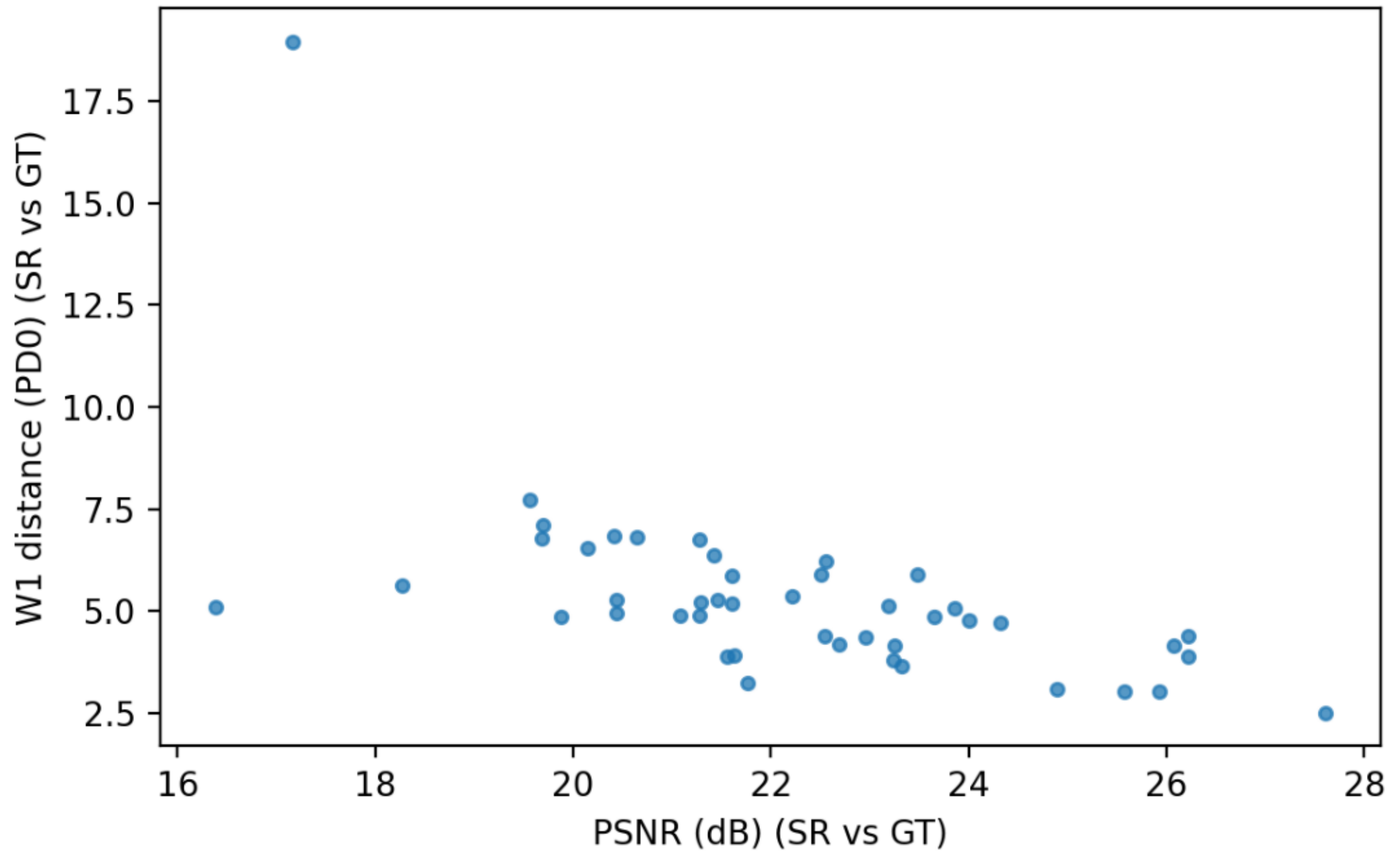
Mean W2(PD1) audit subset

Mean W2(PD1) vs GT on speed (AUDIT subset)



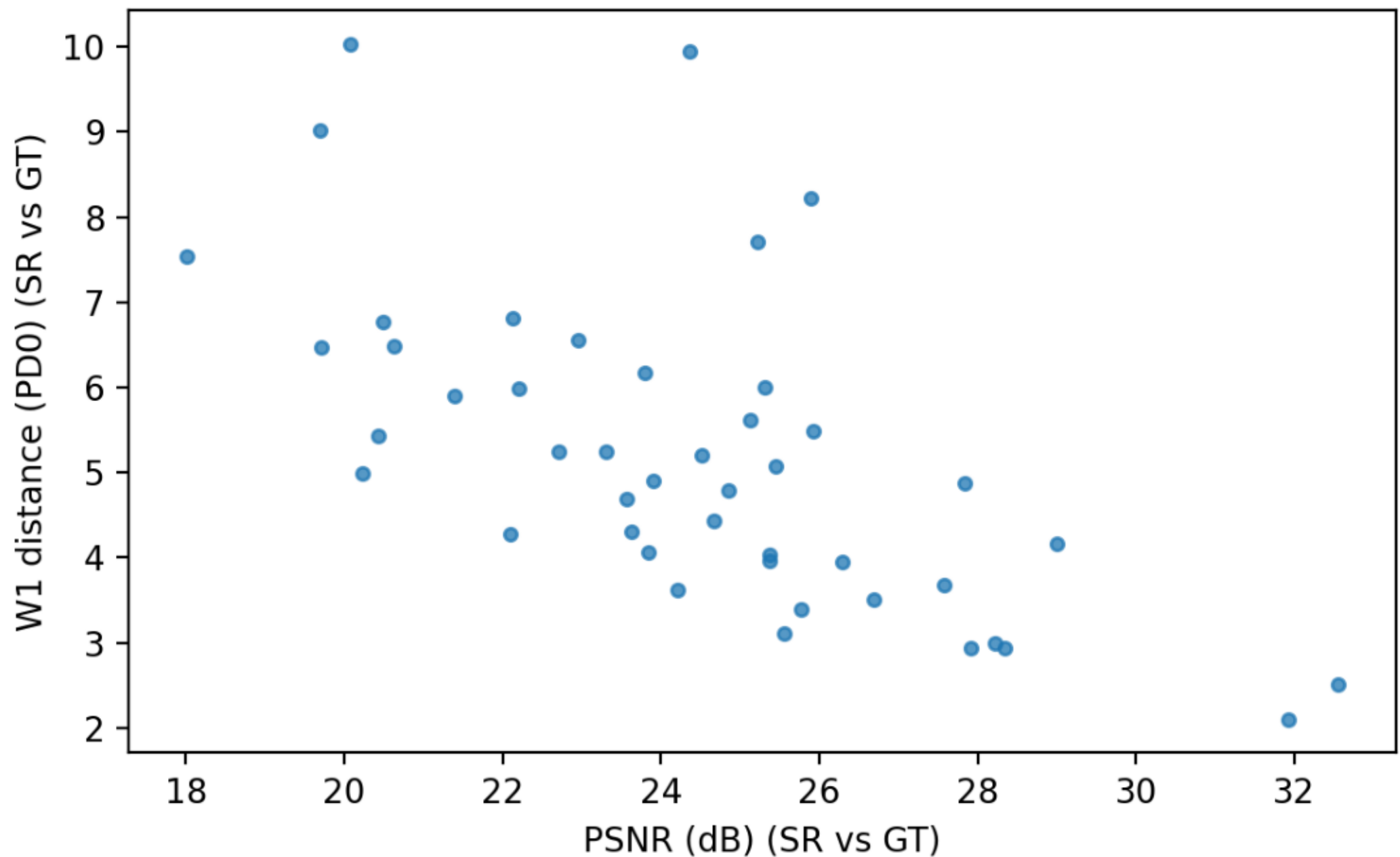
Scatter: PSNR vs W1(PD0) GAN

GAN_MRHR: PSNR vs W1(PD0) on speed (per-patch, n=45)



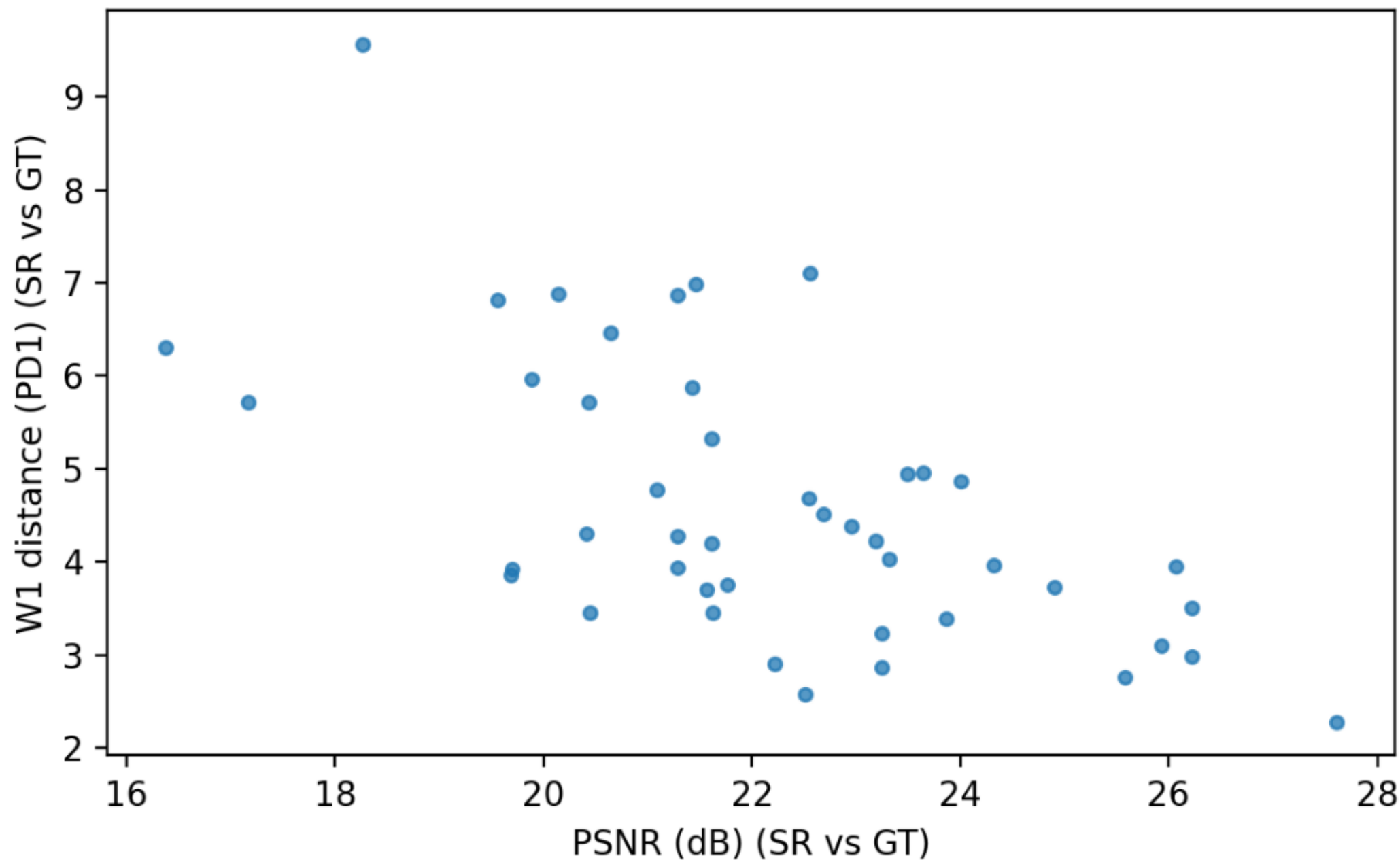
Scatter: PSNR vs W1(PD0) CNN

CNN_MRHR: PSNR vs W1(PD0) on speed (per-patch, n=45)



Scatter: PSNR vs W1(PD1) GAN

GAN_MRHR: PSNR vs W1(PD1) on speed (per-patch, n=45)



Scatter: PSNR vs W1(PD1) CNN

CNN_MRHR: PSNR vs W1(PD1) on speed (per-patch, n=45)

